



Practice Set – Unit II

Random Variables, Distributions,
Expectation and Variance

STA201 – Theory of Probability and Statistics

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Dr. Lokesh Singh

Department of Mathematics,
Amity University Punjab, Mohali.

lsingh@pb.amity.edu

Instructions

- Attempt all questions.
 - Show all necessary steps and calculations.
 - Topics covered: random variables (discrete & continuous), PMF, PDF, CDF, expectation, and variance.
 - Final answers are provided at the end for self-assessment.
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1. Classify the following random variables as discrete or continuous. Justify your answer.

- (a) The number of defective items in a batch of 50.
- (b) The time taken (in seconds) for a chemical reaction to complete.
- (c) The number of calls arriving at a call centre in one hour.
- (d) The weight of a randomly selected package.
- (e) The number of heads obtained when a coin is tossed 10 times.

2. A discrete random variable X has the following probability mass function:

$$p(x) = \frac{x}{10}, \quad x = 1, 2, 3, 4.$$

- (a) Verify that $p(x)$ is a valid PMF.
- (b) Find $P(X \leq 2)$.
- (c) Find $P(X > 3)$.

3. A random variable X has PMF:

$$p(x) = c(x + 1), \quad x = 0, 1, 2, 3.$$

- (a) Find the value of c .
- (b) Find $P(X \geq 2)$.
- (c) Find $E[X]$.

4. A continuous random variable X has PDF:

$$f(x) = kx^2, \quad 0 \leq x \leq 2.$$

- (a) Find the value of k .
- (b) Find $P(0.5 \leq X \leq 1.5)$.
- (c) Find the CDF $F(x)$.

5. For the PDF $f(x) = 2x$, $0 \leq x \leq 1$:

- (a) Find $P(X \leq 0.5)$.
- (b) Find $P(0.3 \leq X \leq 0.8)$.
- (c) Find $E[X]$ and $\text{Var}(X)$.

6. The CDF of a random variable X is given by:

$$F(x) = \begin{cases} 0, & x < 0, \\ \frac{x^2}{4}, & 0 \leq x \leq 2, \\ 1, & x > 2. \end{cases}$$

- (a) Find the PDF $f(x)$.
- (b) Find $P(0.5 \leq X \leq 1.5)$.
- (c) Find $E[X]$.

7. A fair six-sided die is rolled. Let X be the face value.

- (a) Write the PMF of X .
- (b) Find $E[X]$ and $\text{Var}(X)$.

8. A random variable X has PMF:

$$p(x) = \begin{cases} 0.2, & x = 0, \\ 0.5, & x = 1, \\ 0.3, & x = 2. \end{cases}$$

- (a) Find $E[X]$ and $E[X^2]$.
- (b) Hence find $\text{Var}(X)$.

9. A continuous random variable X has PDF:

$$f(x) = \frac{1}{4}, \quad 0 \leq x \leq 4.$$

- (a) Verify that $f(x)$ is a valid PDF.
- (b) Find $P(1 \leq X \leq 3)$.
- (c) Find $E[X]$ and $\text{Var}(X)$.

10. A random variable X has PDF:

$$f(x) = e^{-x}, \quad x \geq 0.$$

- (a) Verify that $f(x)$ is a valid PDF.
- (b) Find $P(X > 2)$.
- (c) Find $E[X]$ and $\text{Var}(X)$.

11. The time T (in hours) until failure of a component has PDF:

$$f(t) = \lambda e^{-\lambda t}, \quad t \geq 0.$$

Given that the expected lifetime is 500 hours, find:

- (a) the value of λ ,
- (b) $P(T > 600)$,

(c) $\text{Var}(T)$.

12. A discrete random variable X has PMF:

$$p(x) = \frac{1}{5}, \quad x = 1, 2, 3, 4, 5.$$

Find:

- (a) $E[X]$,
- (b) $E[X^2]$,
- (c) $\text{Var}(X)$.

13. A continuous random variable X has PDF:

$$f(x) = \begin{cases} x, & 0 \leq x \leq 1, \\ 2 - x, & 1 < x \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Verify that $f(x)$ is a valid PDF.
- (b) Find $P(0.5 \leq X \leq 1.5)$.
- (c) Find $E[X]$.

14. If $E[X] = 5$ and $\text{Var}(X) = 4$, find:

- (a) $E[3X + 2]$,
- (b) $\text{Var}(3X + 2)$,
- (c) $E[X^2]$.

15. A random variable X has $E[X] = 10$ and $E[X^2] = 120$. Find:

- (a) $\text{Var}(X)$,
- (b) the standard deviation of X ,
- (c) $\text{Var}(2X - 5)$.

16. A box contains 5 items, 2 of which are defective. Two items are drawn without replacement. Let X be the number of defective items drawn.

- (a) Find the PMF of X .
- (b) Find $E[X]$.
- (c) Find $\text{Var}(X)$.

17. The PDF of a random variable X is:

$$f(x) = c(1 - x^2), \quad -1 \leq x \leq 1.$$

- (a) Find c .
- (b) Find $E[X]$.
- (c) Find $\text{Var}(X)$.

18. A random variable X has the following CDF:

$$F(x) = \begin{cases} 0, & x < 0, \\ 1 - e^{-2x}, & x \geq 0. \end{cases}$$

- (a) Find the PDF $f(x)$.
- (b) Find $P(1 \leq X \leq 2)$.
- (c) Find $E[X]$.

19. A discrete random variable X has PMF:

$$p(x) = \frac{k}{x}, \quad x = 1, 2, 3, 4.$$

- (a) Find k .
- (b) Find $P(X \leq 3)$.
- (c) Find $E[X]$.

20. A continuous random variable X has PDF:

$$f(x) = 3x^2, \quad 0 \leq x \leq 1.$$

- (a) Find the CDF $F(x)$.
- (b) Find $P(X \leq 0.5)$.
- (c) Find $E[X]$, $E[X^2]$, and $\text{Var}(X)$.